

Conflicts of Interest (COI) Statement

Building a Unified Science of Natural Behavior
with Densely Overlapping Measurement Environments (Domes)

FreeMoCap Foundation, Inc. — Written Proposal to the NSF X-Labs Initiative

Lead Organization: FreeMoCap Foundation, Inc. · Principal Investigator: Jonathan Matthis, PhD

1. SUMMARY

FreeMoCap Foundation, Inc. (the “Foundation”), the lead organization, and its Senior/Key Personnel have reviewed their financial interests and outside affiliations against the scope of the proposed work in accordance with §10.2 of the solicitation and, as a framework, FAR Subpart 9.5. **We have identified no actual or potential organizational or personal conflict of interest that would impair the objectivity of the proposed work or confer an unfair competitive advantage.** The Foundation is a nonprofit whose flagship technology is released free and open-source; it holds no proprietary product, patent, license, or equity position whose value depends on the outcome of this award. Several Senior/Key Personnel maintain concurrent outside employment in unrelated industries; these affiliations are disclosed below in the interest of full transparency, are unrelated to the measurement and instrumentation domain of the proposed DOME platform, and are being wound down as personnel transition to full-time roles across Phase 0 and Phase 1. The management measures in §4 and the ongoing-disclosure commitment in §5 govern any conflict that may arise.

2. ORGANIZATIONAL INTERESTS — FREEMOCAP FOUNDATION, INC.

The Foundation is a mission-driven nonprofit corporation. Its core technology, the FreeMoCap software platform, is distributed at no cost under an open-source license. The Foundation therefore has **no financial interest in any particular scientific or commercial outcome** of the proposed work: it does not sell the software, does not hold patents or exclusive licenses tied to the DOME instrument, and does not hold equity in, or receive royalties from, any entity positioned to benefit from the award.

The Foundation has no other federal contract, grant, or advisory relationship that would create an organizational conflict of interest under FAR Subpart 9.5 — no impaired-objectivity conflict (it has not evaluated, set requirements for, or advised NSF on this program), no unequal-access-to-information conflict, and no biased-ground-rules conflict. The Foundation has no current or planned subcontractor, consultant, or vendor arrangement in which a Senior/Key Personnel member or their affiliated organization would be positioned to profit from the proposed work. Should any such arrangement be contemplated during performance, it will be disclosed and resolved in advance under §5.

3. SENIOR/KEY PERSONNEL — FINANCIAL INTERESTS AND OUTSIDE AFFILIATIONS

Each individual below has disclosed outside employment and any financial interest relevant to the proposed work. Personnel not listed with an outside affiliation report no relevant outside financial interest. None of the outside affiliations operate in the scientific-instrumentation, motion-capture, or sensing-and-imaging domain of the proposed DOME platform; none holds a supplier, subcontractor, or competitive relationship to the proposed work.

Individual / Role	Foundation status	Concurrent outside affiliation	Conflict assessment
Jonathan Matthis, PhD President / PI	Full-time	None. Former tenure-track faculty, Northeastern University (resigned). NIH NEI K99/R00 awardee (EY028229), now concluded.	No conflict. No outside employment or financial interest tied to the award.
AC, PhD Chief Scientific Officer	Full-time	None reported.	No conflict.
NR Chief Executive Officer	Transitioning to full-time by Phase 1	Software engineering (Rapid7); co-founder, MailLift (API startup). Unrelated industries.	No conflict. Outside roles are in general software; no overlap with DOME scope; winds down on transition.
EI Chief Technology Officer	Part-time; transitioning to full-time by Phase 1	Concurrent CTO, Chorus Innovations (enterprise software / health-tech platform). Former Principal Architect, Unqork.	No conflict. Enterprise-software domain, not instrumentation; no supplier/competitor relationship; time commitment managed under §4.
JKL, PhD Chief AI Officer	Transitioning to full-time by Phase 1	Applied AI Architect, SOLID Inc. (applied-AI consulting).	No conflict. Unrelated applied-AI work; no overlap with DOME scope.
RR Chief Financial Officer / Treasurer	Part-time (volunteer CFO)	Founder, Capital Bookkeeping Cooperative; Controller, Honest Weight Food Co-op; Adjunct Professor of Accounting, University at Albany.	No conflict. Accounting/finance and academic roles unrelated to the technical scope; no financial interest in project outcomes.
KM, PhD Project Manager, DOME-Mobile	Transitioning to full-time by Phase 1	Senior Data Scientist, SynMax (energy-sector data analytics).	No conflict. Energy-analytics employer is outside the instrumentation domain; no supplier/competitor relationship.
MN Project Manager, DOME-L	Transitioning to full-time by Phase 1	None reported.	No conflict.

4. AVOIDANCE, NEUTRALIZATION, AND MITIGATION

Although no actual or potential conflict has been identified, the Foundation will maintain the following measures to prevent, neutralize, or mitigate any conflict that could arise during performance, consistent with §10.2 and FAR Subpart 9.5:

- 1. Full-time transition.** The X-Labs model treats Phase 0 as the runway for Senior/Key Personnel to transition to full-time before Phase 1. The PI and Chief Scientific Officer are already full-time; the remaining Senior/Key Personnel commit to full-time by Phase 1, retiring the concurrent outside employment disclosed in §3 and removing any residual divided-commitment concern.
- 2. Scope separation and recusal.** Outside employment is confined to work unrelated to the DOME platform. No Senior/Key Personnel will direct Foundation procurement toward, or

accept Foundation-funded work from, an entity in which they hold a financial interest. Any individual with a personal financial interest in a matter before the Foundation will recuse themselves from Foundation decisions on that matter.

3. **Open, nonprofit governance.** The Foundation's open-source, nonprofit structure and cooperative-style governance keep the technology in the public domain and remove the profit motive that ordinarily drives instrumentation conflicts. Financial oversight is exercised by the CFO/Treasurer and the Foundation's board.
4. **Written COI policy and annual disclosure.** The Foundation will maintain a written conflict-of-interest policy and collect an annual (and event-driven) disclosure from every Senior/Key Personnel member and employee, with review by the board or its designee. Current-and-pending support and biosketches will be provided for all Senior/Key Personnel at the oral-proposal stage.

5. ONGOING DISCLOSURE COMMITMENT

This assessment reflects the facts known as of the submission date. The requirements of §10.2 apply to the Foundation, its employees, and any subcontractors or consultants proposed to perform work. If, at any time — including after award — the Foundation or any subcontractor identifies an actual or potential organizational conflict of interest (using FAR Subpart 9.5 as a framework) that has not been adequately disclosed and resolved, or waived in writing by NSF, the Foundation will promptly provide full written disclosure to NSF describing the conflict and the actions taken or proposed to resolve it, together with a proposed mitigation plan. The Foundation flows this same obligation down to any subcontractor or consultant it engages.

Submitted on behalf of FreeMoCap Foundation, Inc.
Jonathan Matthis, PhD — President and Principal Investigator
Date: July 14, 2026